VOXXEDDAYS BRUSSELS

VoxxedDays Brussels 2023

The Hitchhiker's guide to Software Architecture Design

Raphaël SEMETEYS - Alexandre TOURET

Digital Payments for a Trusted World

WORLDLINE

Who are we?

Raphaël SEMETEYS

Software Architect



@RaphaelSemeteys

blog.worldline.tech

Alexandre TOURET

Software Architect



@touret_alex

blog.touret.info

Who are we?

We design payments technology that powers the growth of millions of businesses around the world.

@worldlinetech

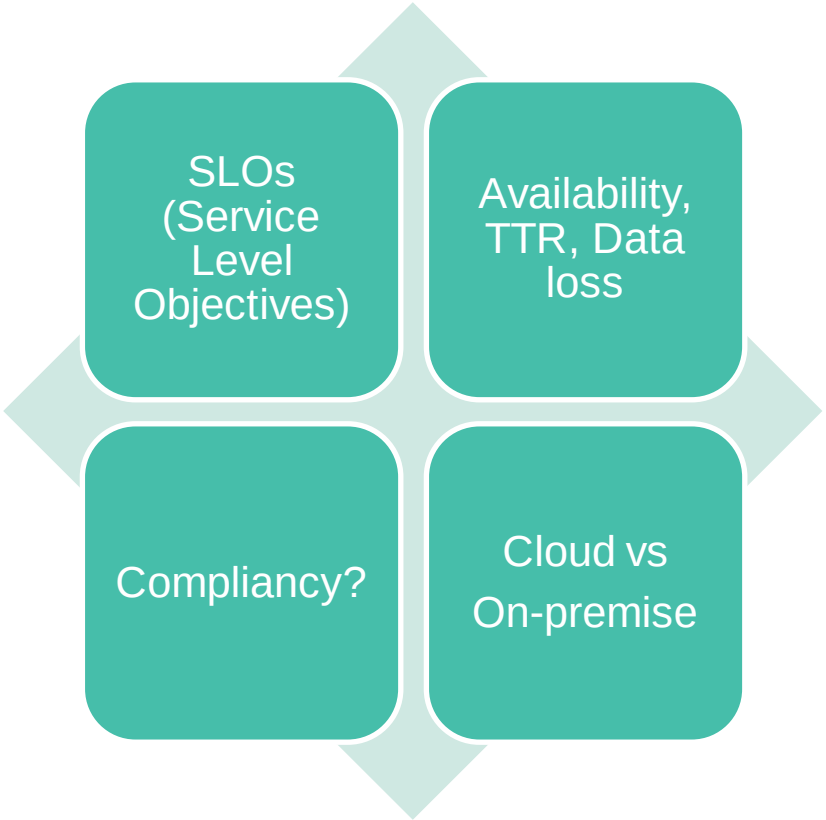
WORLDLINE



Business View



Requirements



Architecture Decision Record

Fill and store every key architectural decision as close as possible to the code

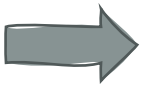
```
# Title

# Status
- [ ] proposed
- [X] accepted
- [ ] rejected
- [ ] deprecated
- [ ] superseded

# Context

# Decision

# Consequences
```



```
# ADR01 – Cloud hosting

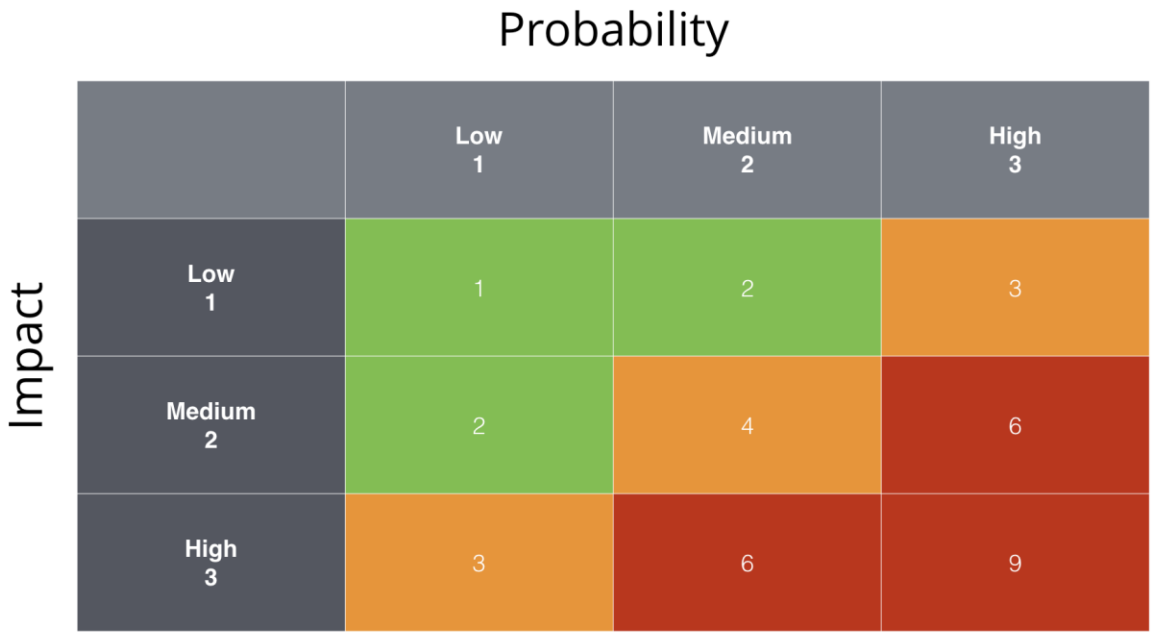
# Status
- [X] proposed
- [ ] accepted
- [ ] rejected
- [ ] deprecated
- [ ] superseded

# Context
Capacity of the platform capacity must adapt according
to Donut@Home success

# Decision
Cloud hosting to optimize pay-per-use and intrinsic
scalability

# Consequences
Cloud-native architecture (12 factors)
```

Risk Analysis?



Source: riskstorming.com

Probability

Impact

	Low 1	Medium 2	High 3
Low 1			
Medium 2			
High 3	- Unavailability of middleware - Database access error - SAN error - VLAN network error		

Business view wrap-up

User needs

- Donut design and delivery application

Requirements

- Retrieved Time Objective, Recovery Point Objective: ?
- TTR: 90% of the transactions up to 2sec
- Availability: 95%
- Number of users: 500 000 / day → *Currently a little bit foggy*
- Ability to easily integrate new features

Compliancy

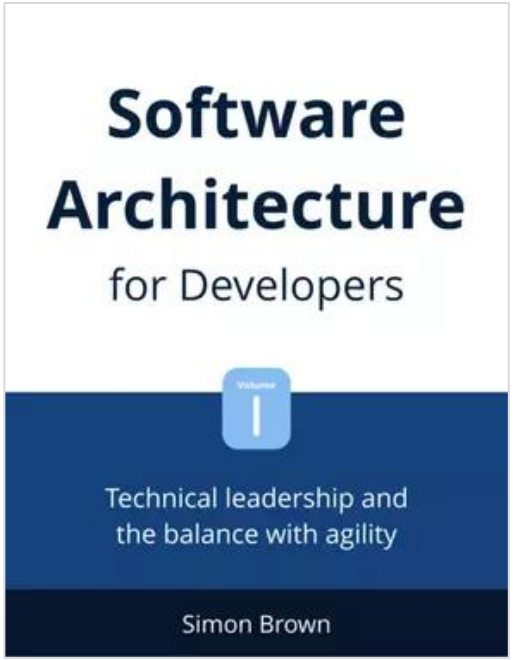
- Compliant with banking and payment standards
- GDPR compliant

Last but not least

- Track every decision into ADRs
- Look into and write risks to deal with (or not)

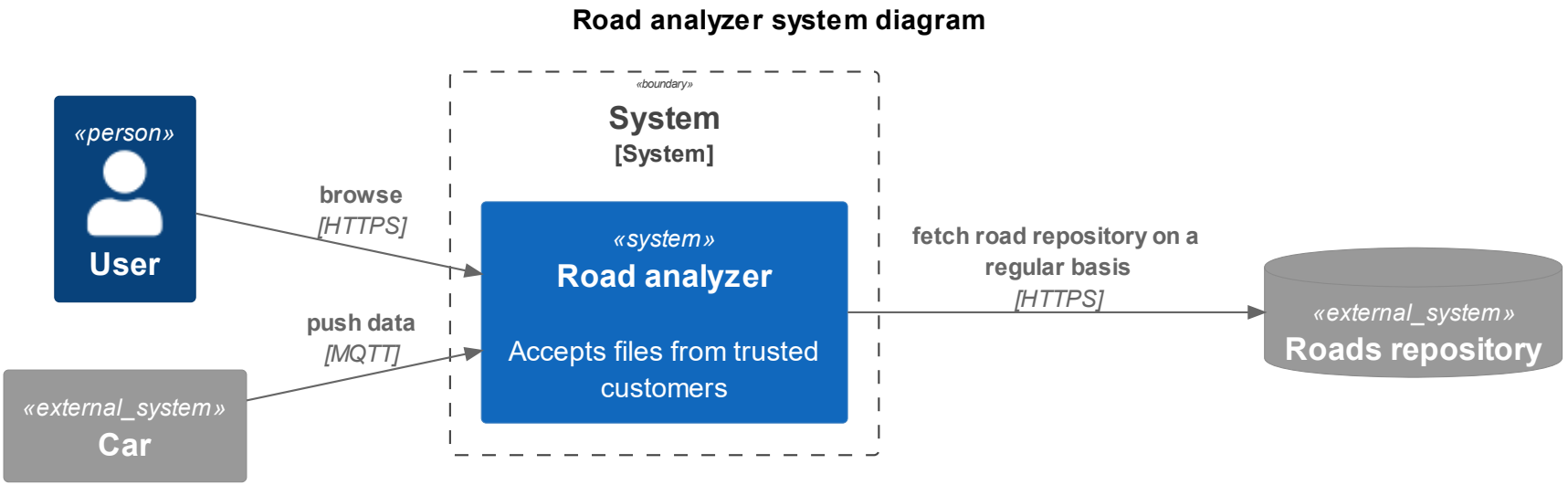
Architecture Approach

C4

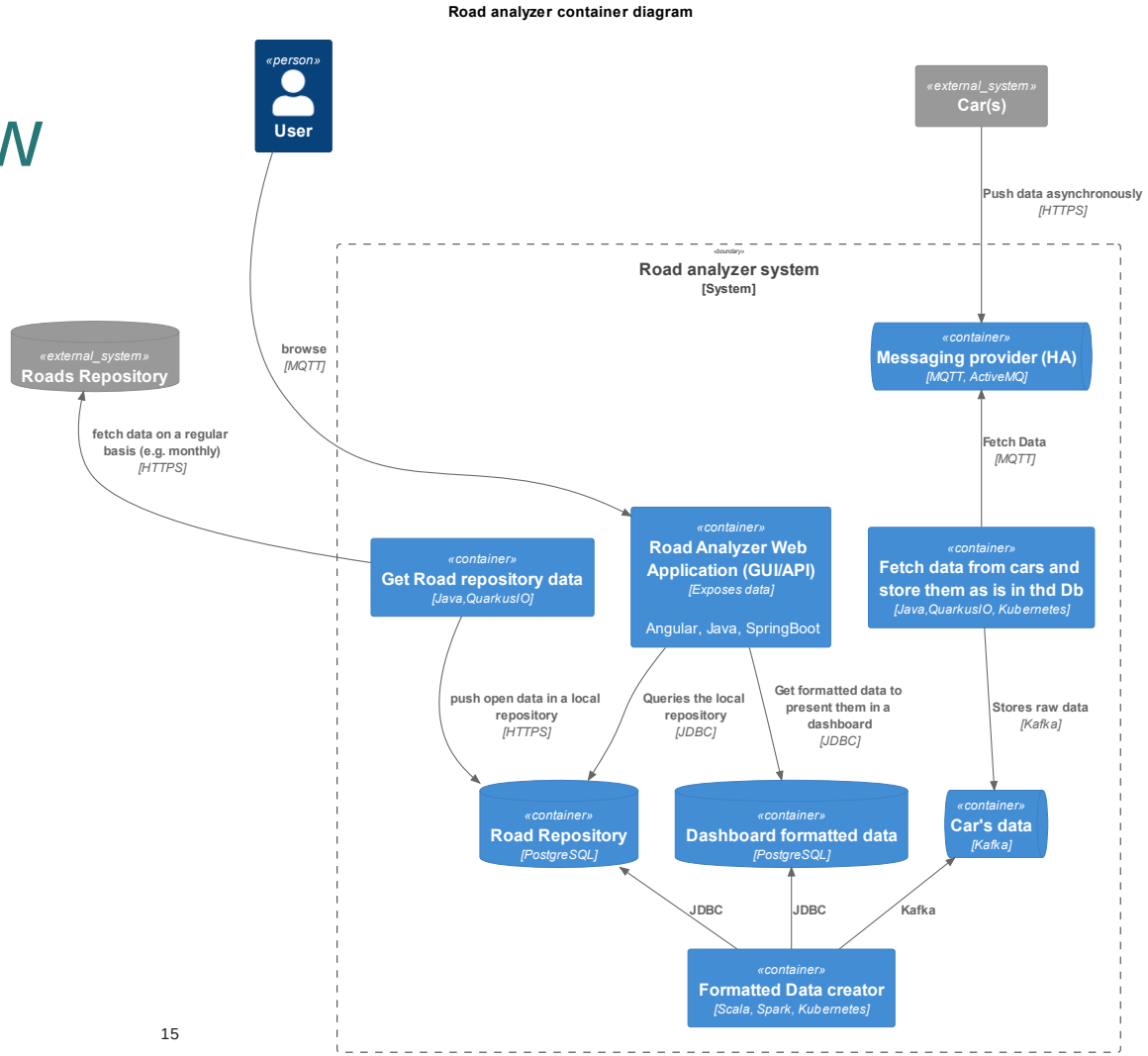


c4model.com

C4 System View

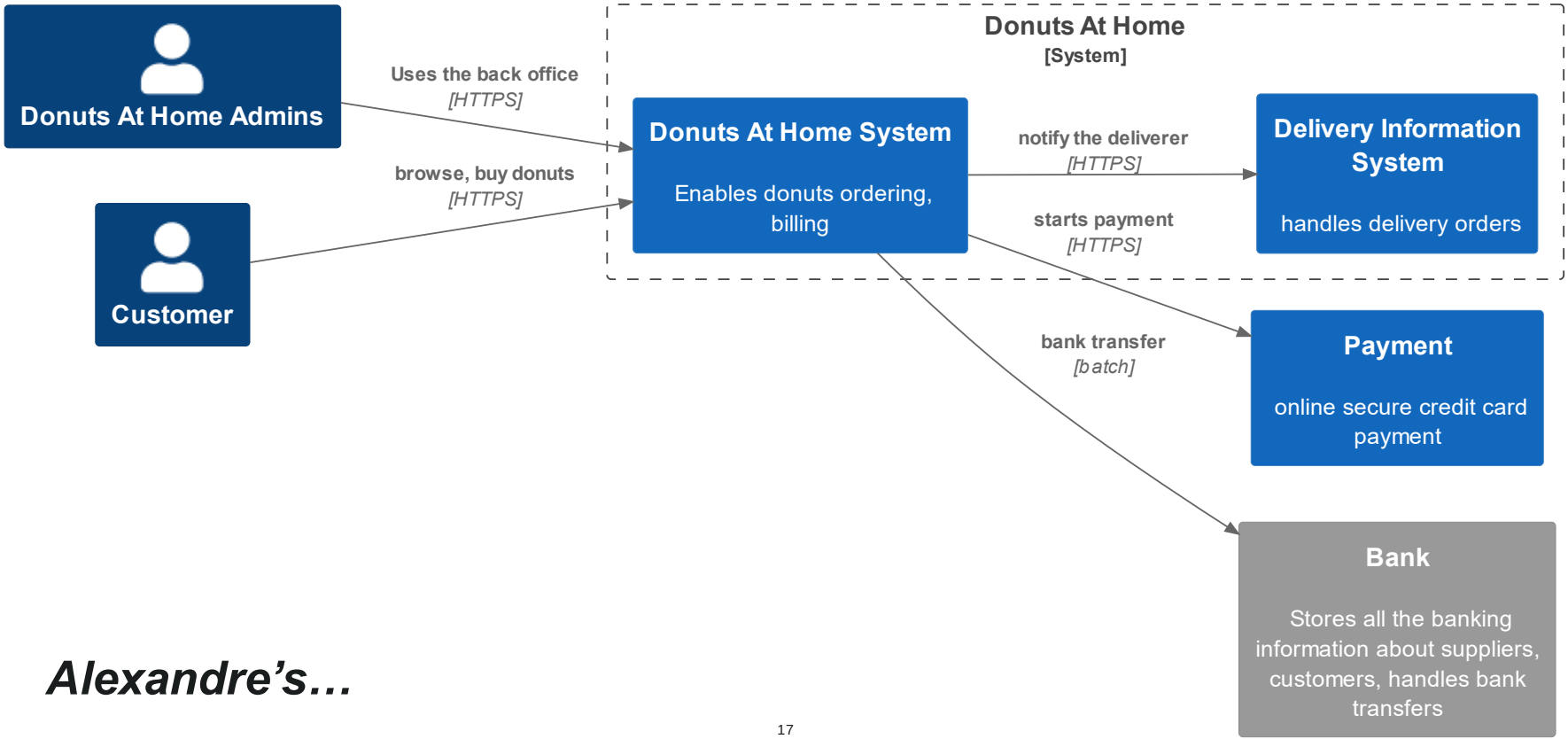


C4 Container View



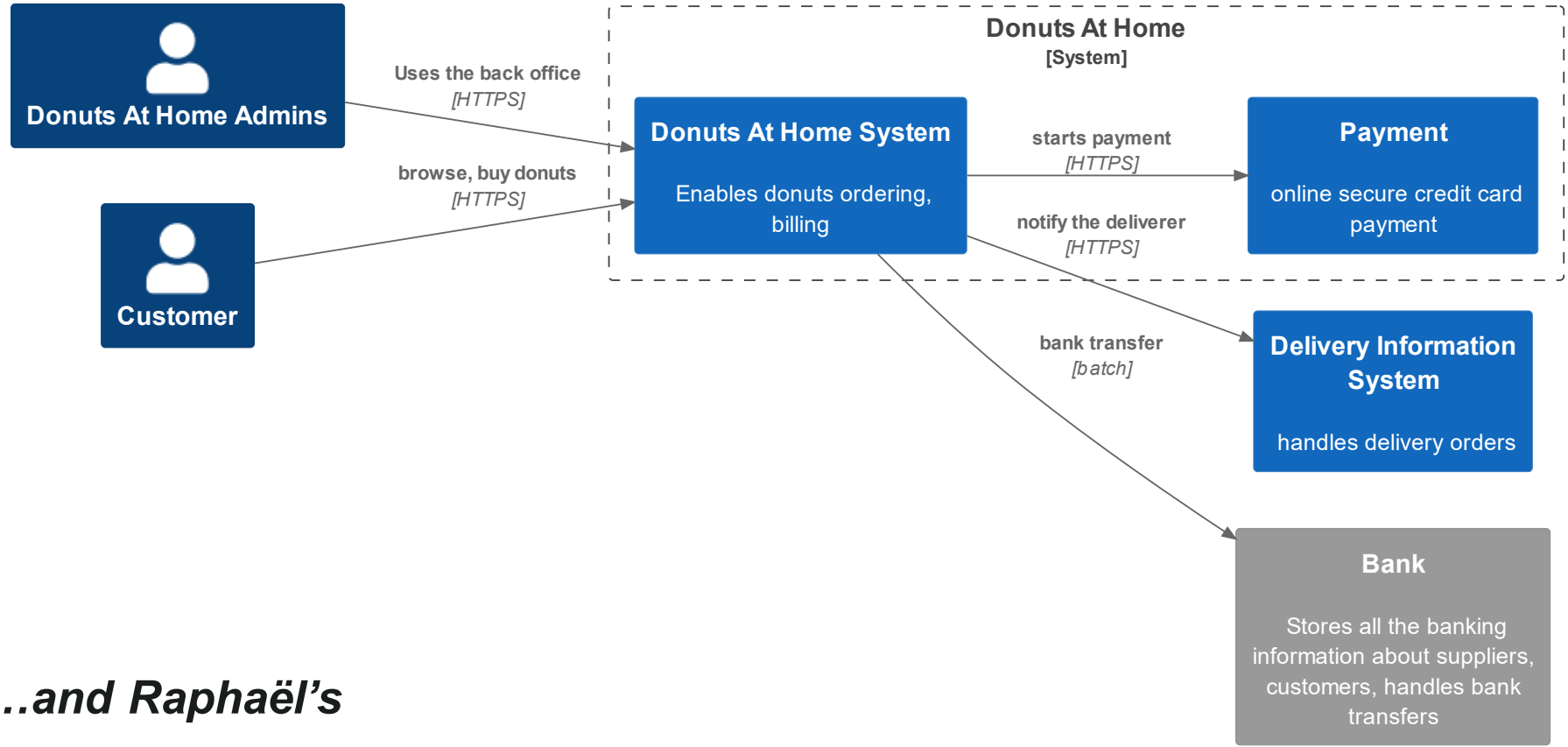
Functional View

Functional view



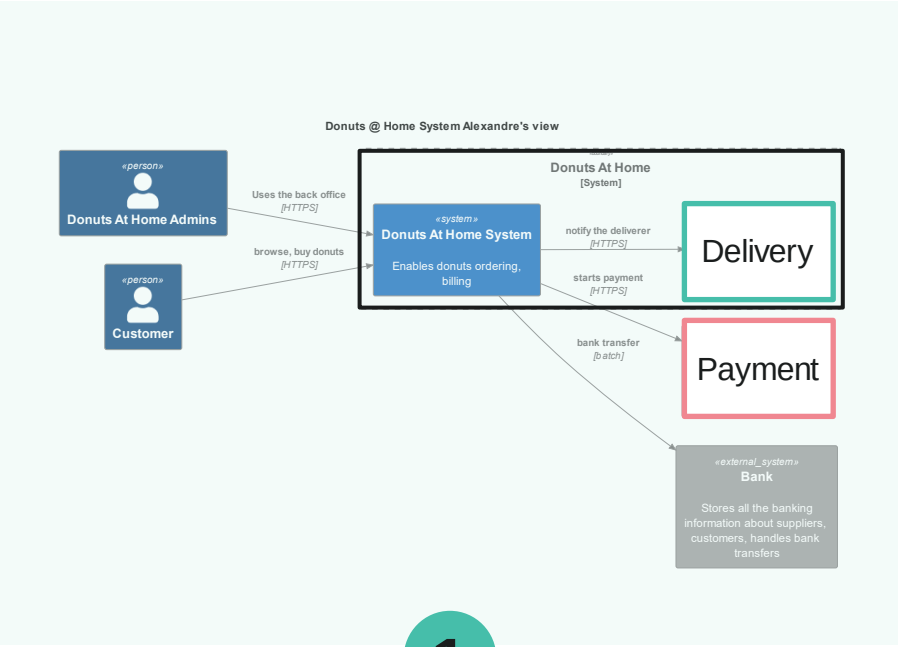
Alexandre's…

Functional view



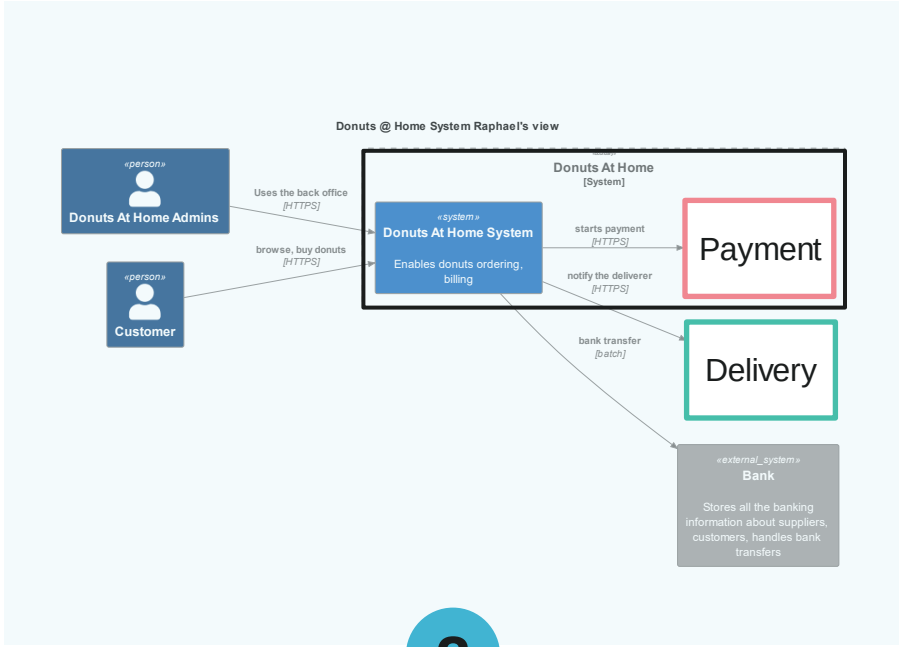
…and Raphaël's

In your opinion?



1

Alexandre's view?

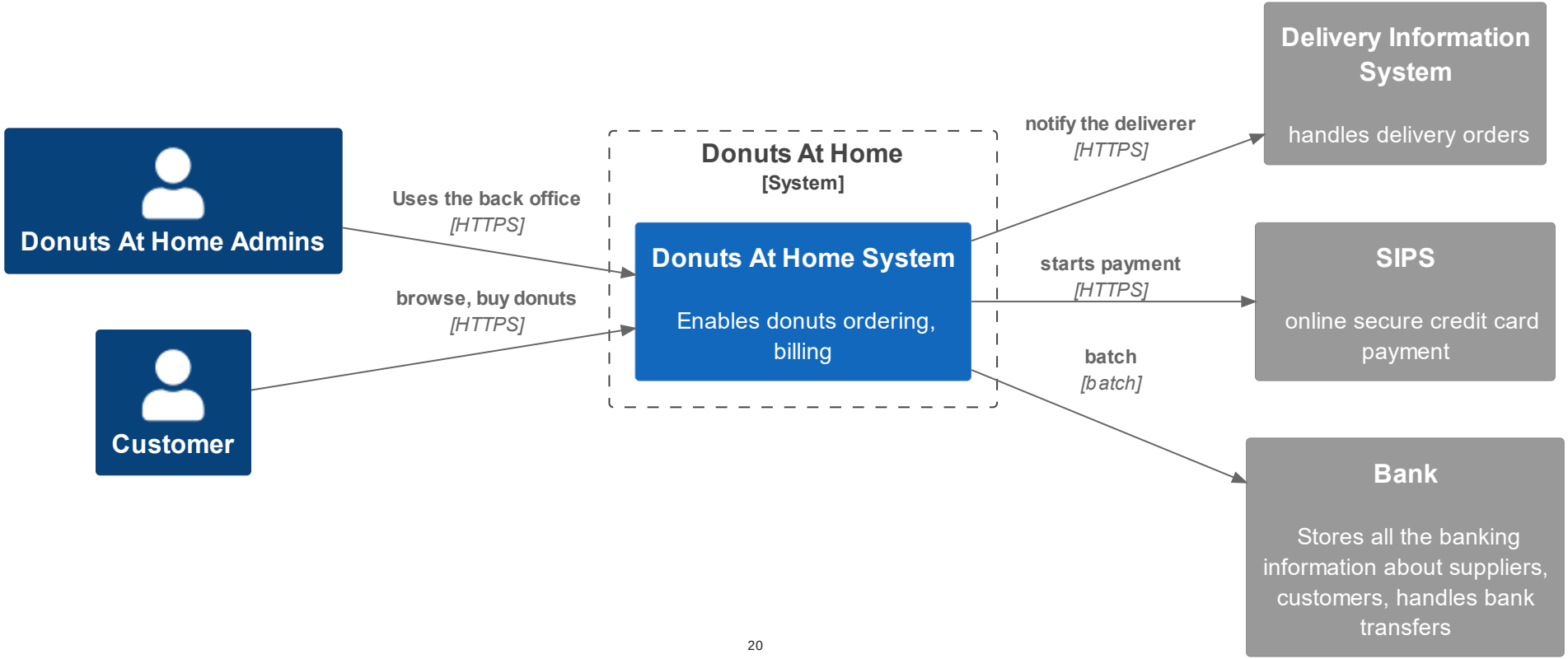


2

Or Raphaël's view?

Functional view

In the end



To sum-up



Design the architecture through several views



Confront different viewpoints



Avoid
« Not Invented Here »

Application View

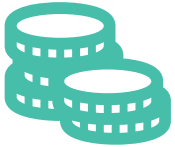
Main characteristics of an architecture



Evolutivity



Modularity



Cost



Performance



Simplicity

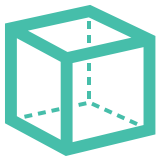


Testability



Fault tolerance

Types of architecture



Monolith



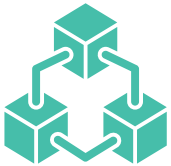
SOA



Orchestration

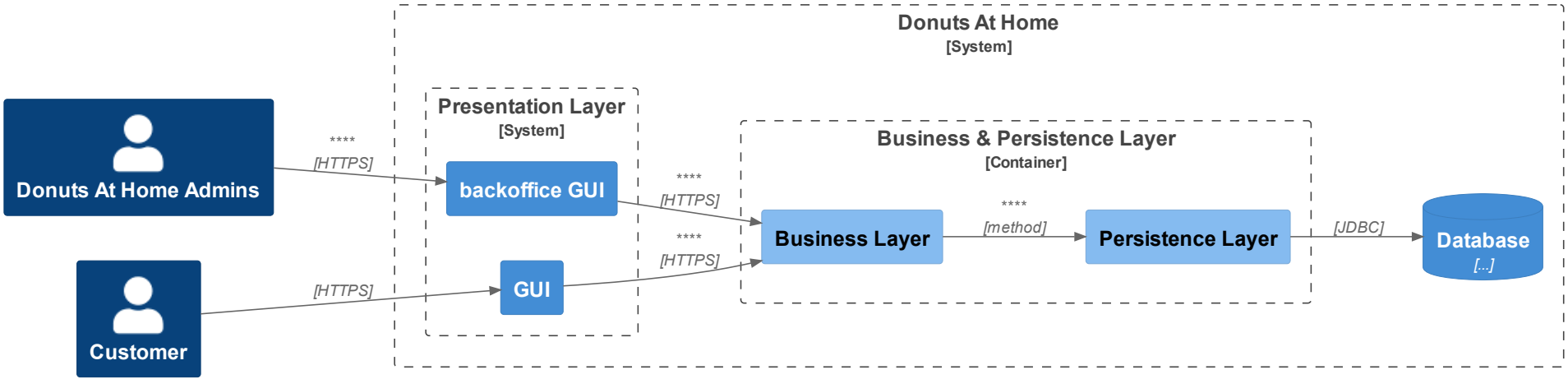


Event Driven

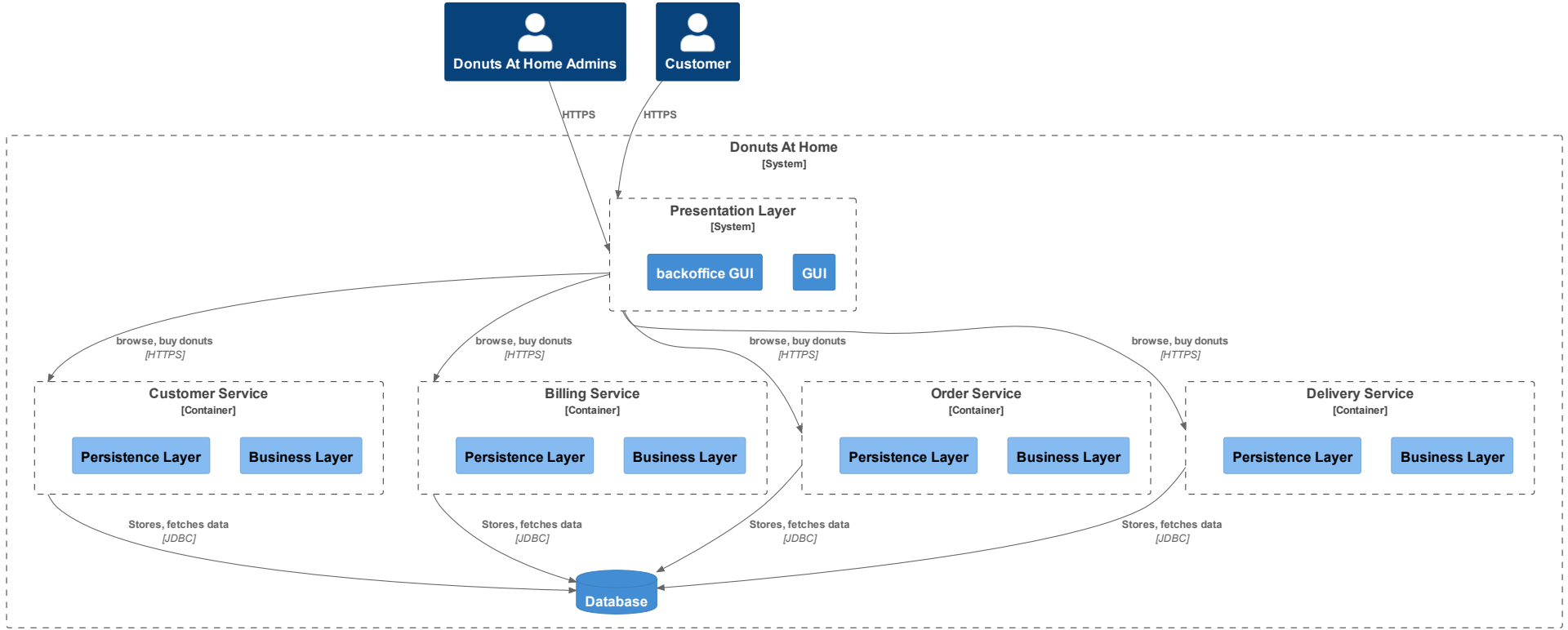


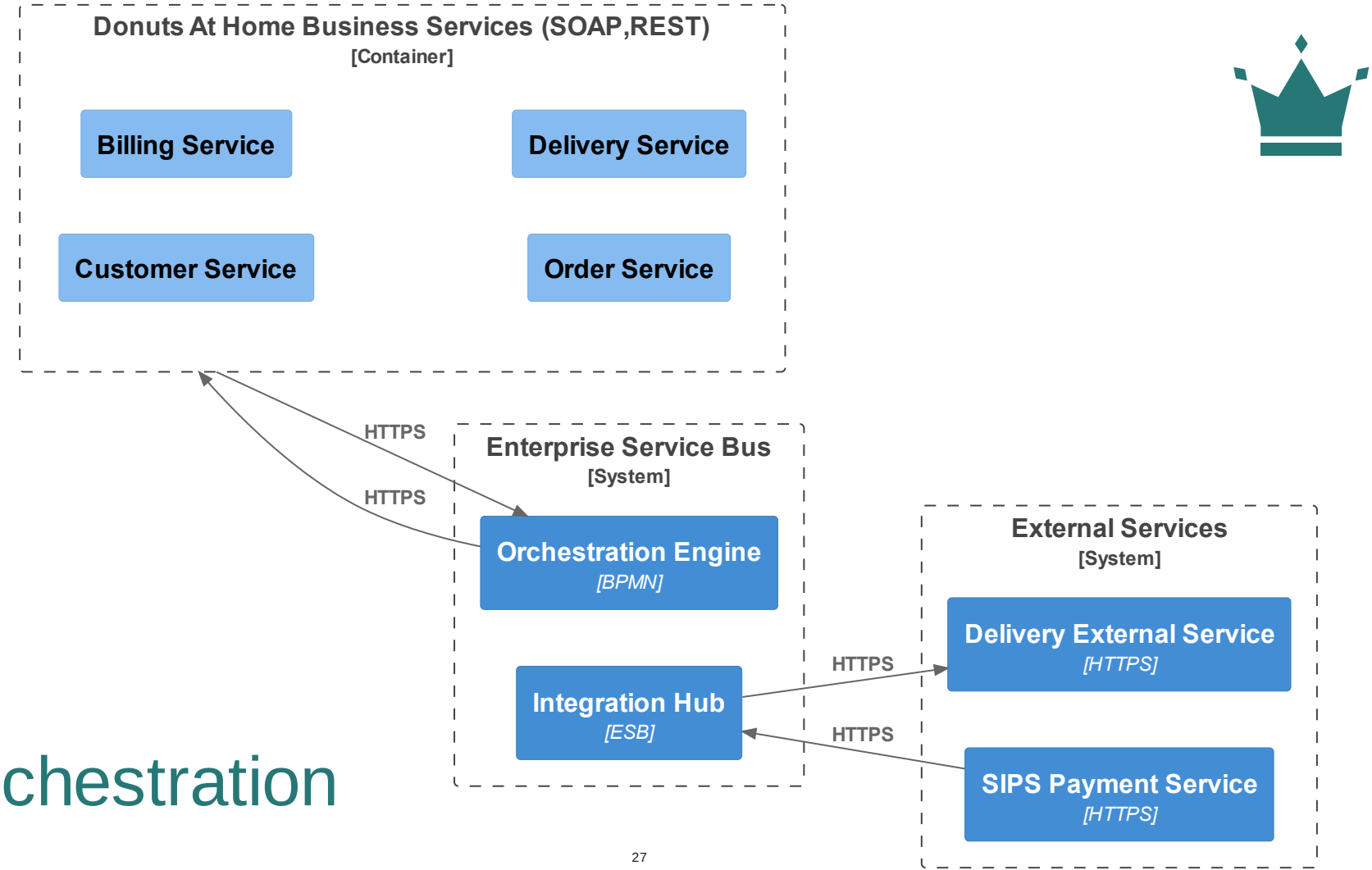
Micro services

Monolith


Donuts At Home
[System]
Presentation Layer
[System]
Business & Persistence Layer
[Container]
Donuts At Home Admins
Customer
backoffice GUI
GUI
Business Layer
Persistence Layer
Database
[...]
****
[HTTPS]
****
[HTTPS]
****
[HTTPS]
[HTTPS]
****
[method]
[JDBC]

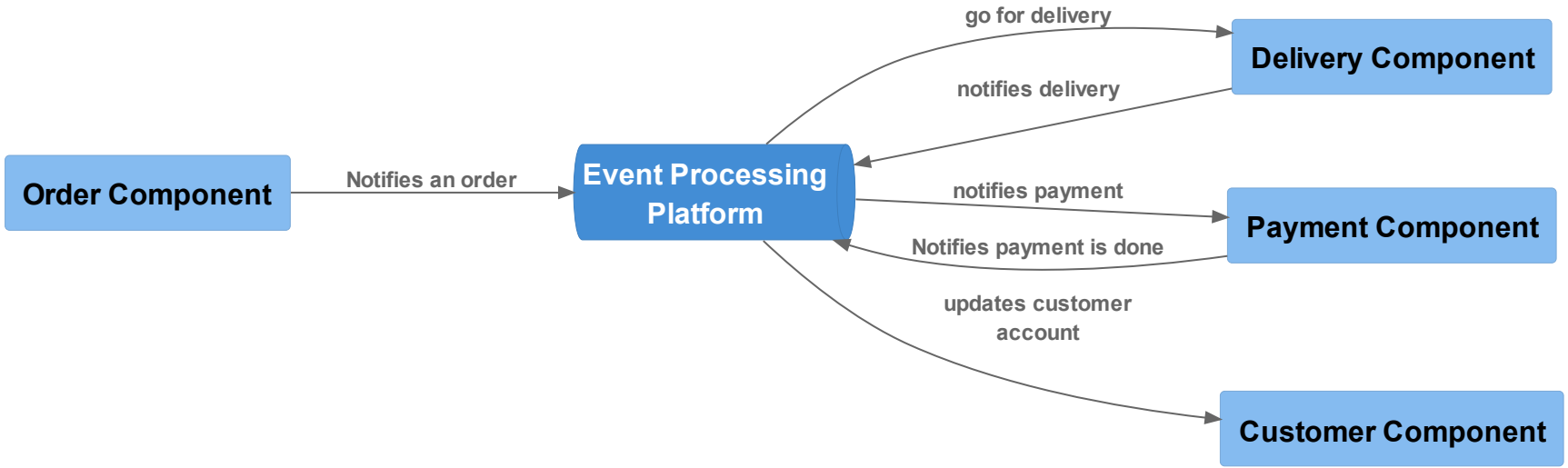
SOA


Donuts At Home Admins
Customer
HTTPS
HTTPS
Donuts At Home
[System]
Presentation Layer
[System]
backoffice GUI
GUI
browse, buy donuts
[HTTPS]
browse, buy donuts
[HTTPS]
browse, buy donuts
[HTTPS]
browse, buy donuts
[HTTPS]
Customer Service
[Container]
Persistence Layer
Business Layer
Billing Service
[Container]
Persistence Layer
Business Layer
Order Service
[Container]
Persistence Layer
Business Layer
Delivery Service
[Container]
Persistence Layer
Business Layer
Stores, fetches data
[JDBC]
Stores, fetches data
[JDBC]
Stores, fetches data
[JDBC]
Stores, fetches data
[JDBC]
Database

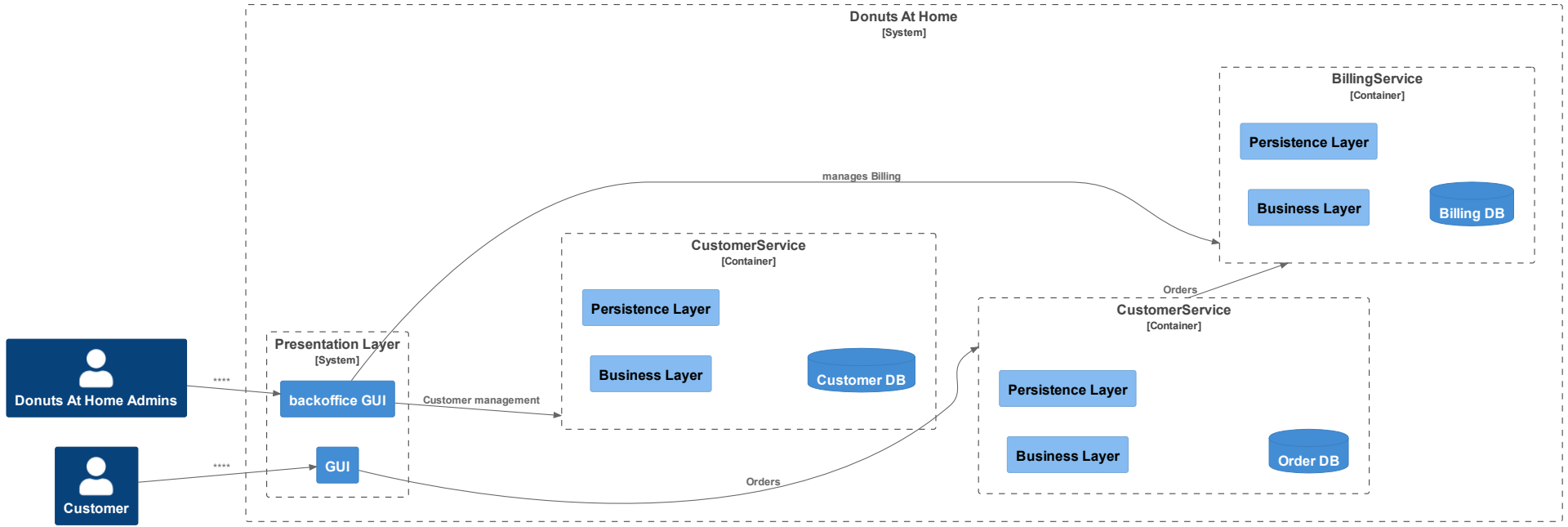


Orchestration

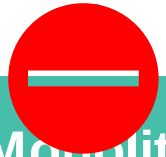
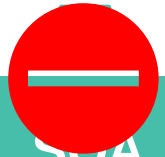
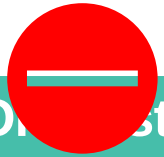
Event Driven



Microservices

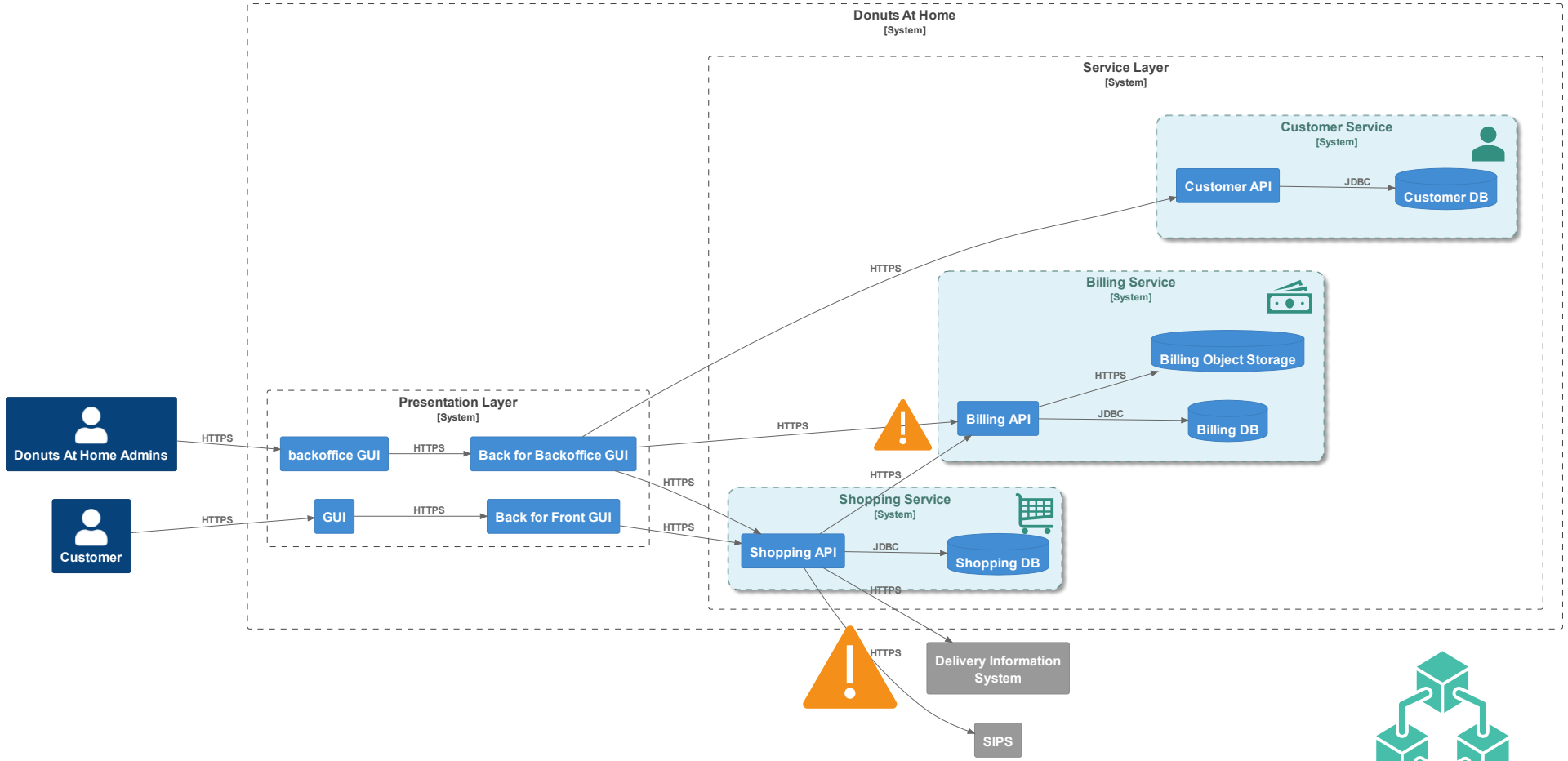


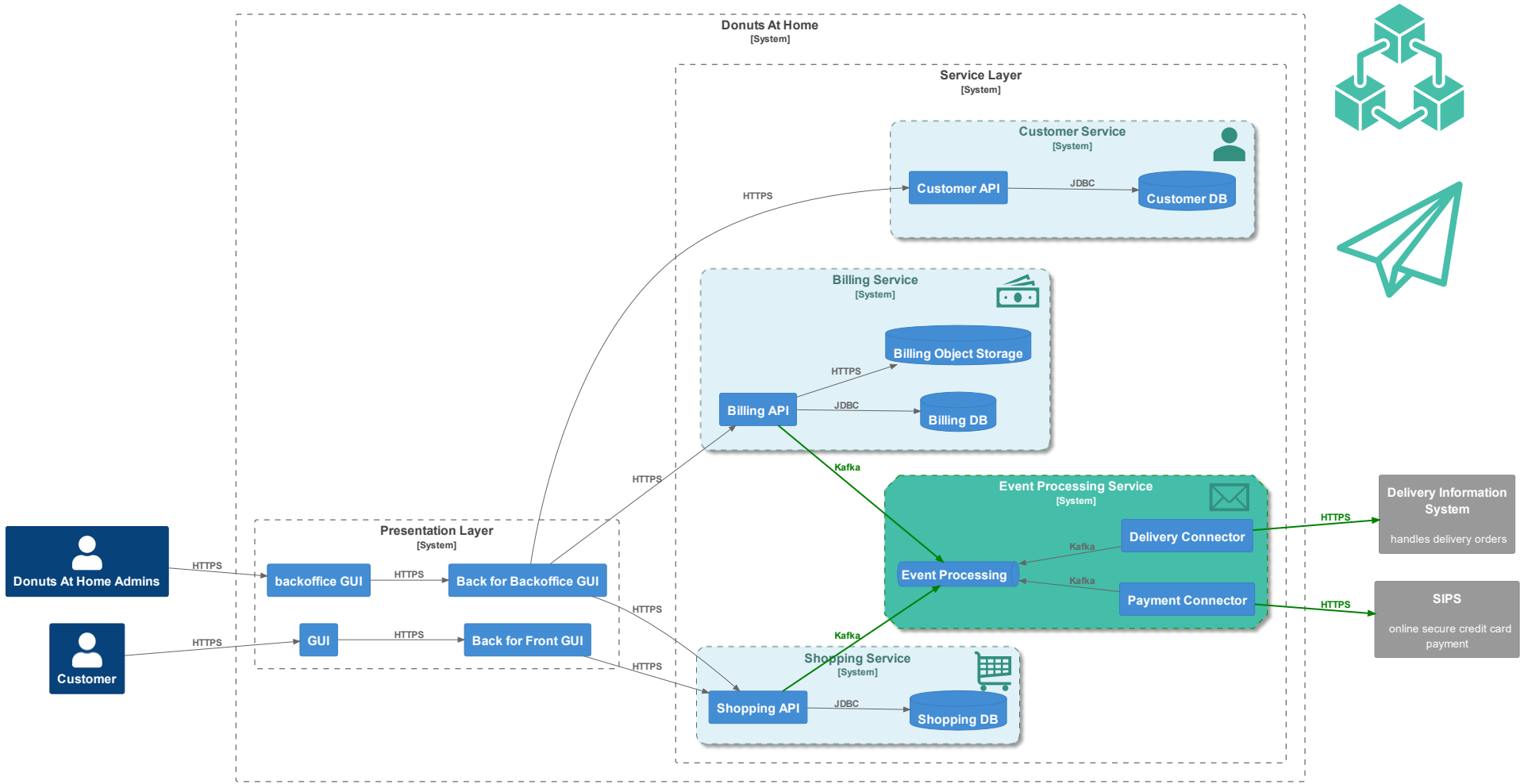
When to use them?

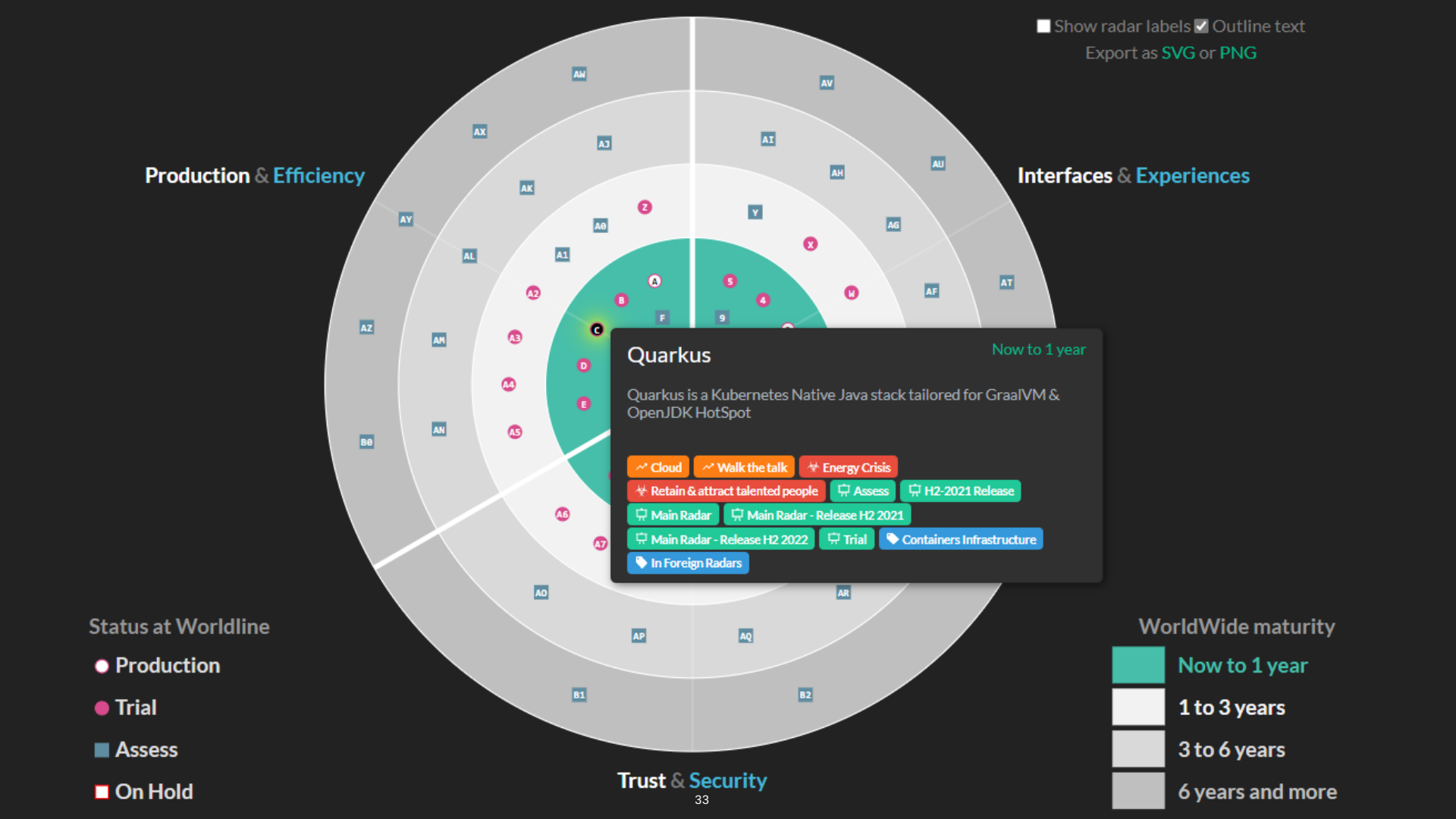




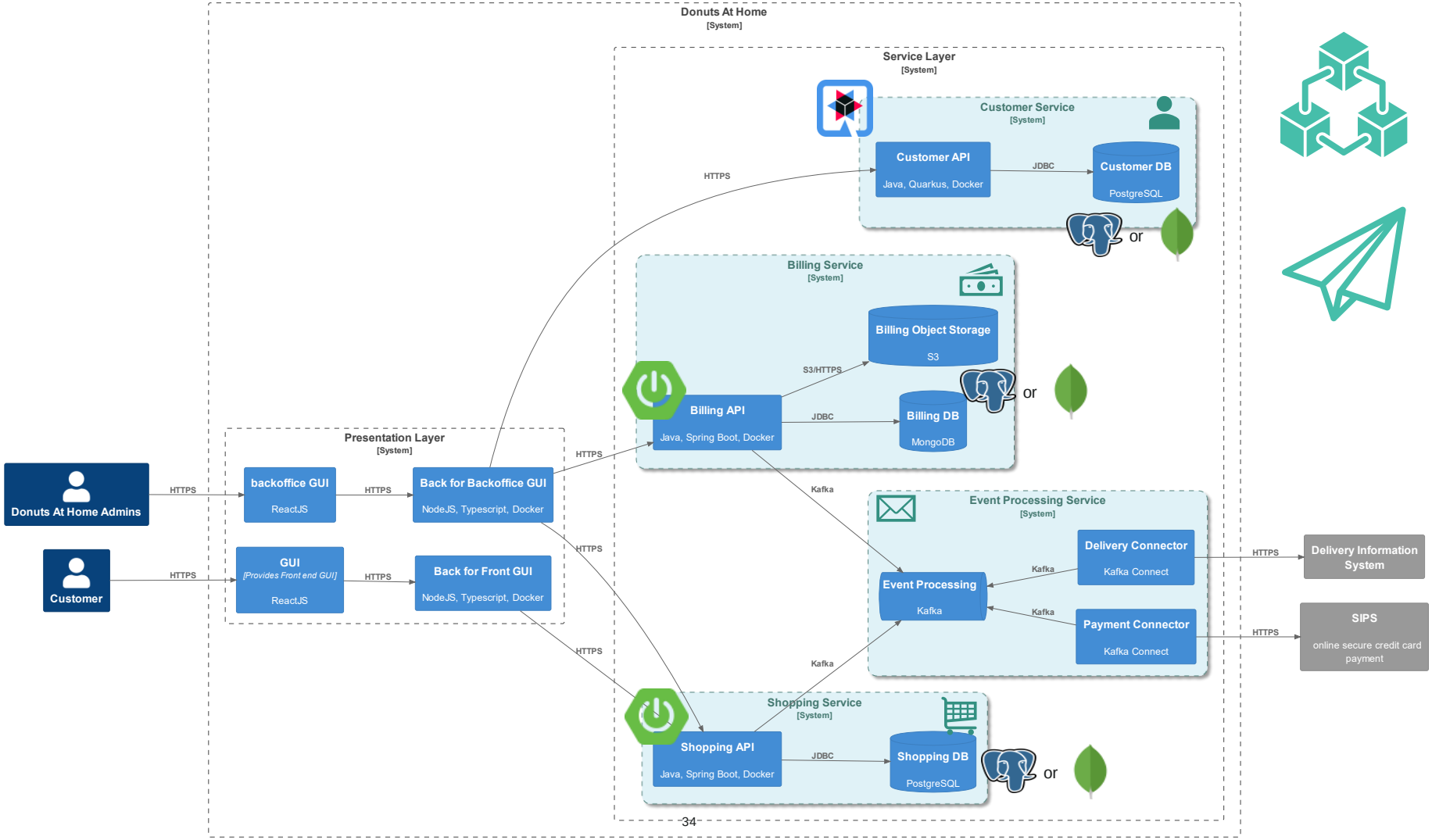


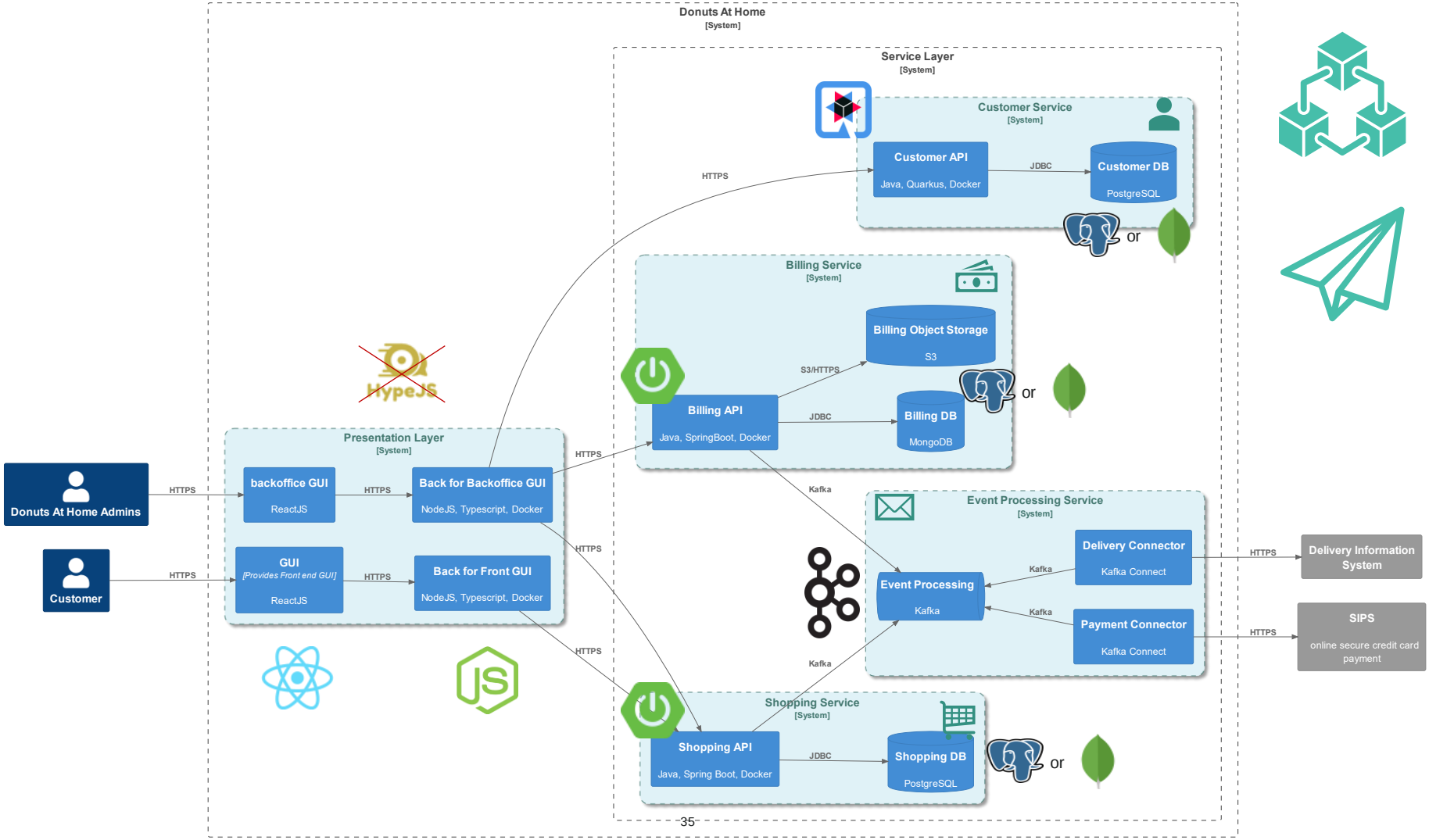
	Monolith	SOA	Orchest-ration	Event Driven	Micro-services
Evolutivity	▲	▲▲▲	▲	▲▲▲▲▲	▲▲▲▲▲
Scalability	▲	▲▲▲	▲▲▲▲	▲▲▲▲▲	▲▲▲▲▲
Modularity	▲	▲▲▲▲	▲▲▲	▲▲▲▲	▲▲▲▲▲
Cost	▲▲▲▲▲	▲▲▲▲	▲	▲▲▲	▲▲▲
Performance	▲▲	▲▲▲	▲▲	▲▲▲▲▲	▲▲
Simplicity	▲▲▲▲▲	▲▲▲	▲	▲	▲
Testability	▲▲	▲▲▲	▲	▲▲	▲▲▲▲
Fault tolerance	▲	▲▲▲▲	▲▲▲	▲▲▲▲▲	▲▲▲▲▲


Donuts At Home
[System]
Service Layer
[System]
Customer Service
[System]
Customer API
JDBC
Customer DB
HTTPS
Billing Service
[System]
Billing Object Storage
HTTPS
Billing API
JDBC
Billing DB
Donuts At Home Admins
HTTPS
Presentation Layer
[System]
backoffice GUI
HTTPS
Back for Backoffice GUI
HTTPS
HTTPS
HTTPS
Shopping Service
[System]
Customer
HTTPS
GUI
HTTPS
Back for Front GUI
HTTPS
Shopping API
JDBC
Shopping DB
HTTPS
HTTPS
Delivery Information System
SIPS


Donuts At Home
[System]
Service Layer
[System]
Customer Service
[System]
Customer API
JDBC
Customer DB
HTTPS
Billing Service
[System]
Billing Object Storage
HTTPS
Billing API
JDBC
Billing DB
Kafka
Event Processing Service
[System]
Delivery Connector
Kafka
Event Processing
Kafka
Payment Connector
HTTPS
Delivery Information System
handles delivery orders
HTTPS
SIPS
online secure credit card payment
Shopping Service
[System]
Shopping API
JDBC
Shopping DB
Kafka
Presentation Layer
[System]
backoffice GUI
HTTPS
Back for Backoffice GUI
GUI
HTTPS
Back for Front GUI
HTTPS
HTTPS
HTTPS
HTTPS
Donuts At Home Admins
HTTPS
Customer
HTTPS

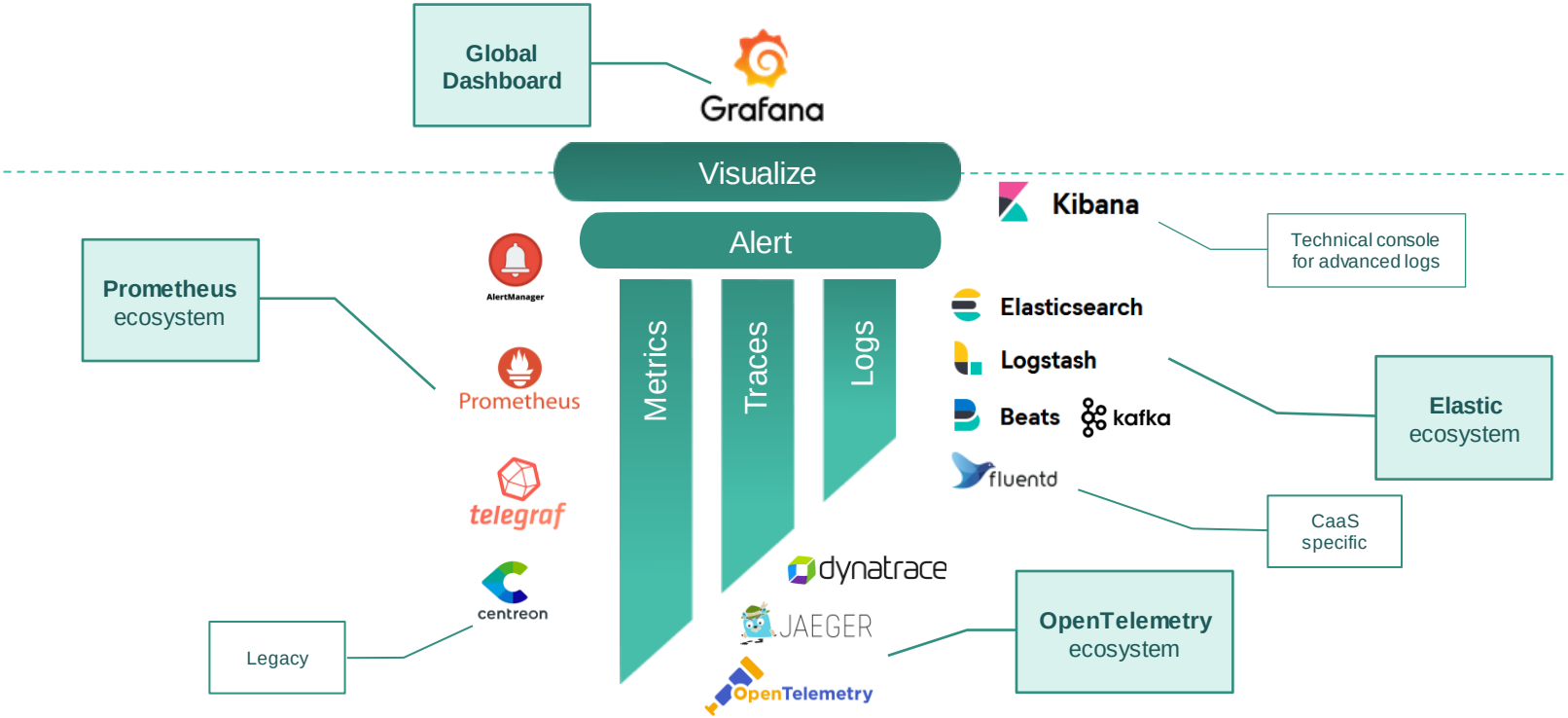

Show radar labels
Outline text
Export as SVG or PNG
Production & Efficiency
Interfaces & Experiences
Trust & Security
Quarkus
Now to 1 year
Quarkus is a Kubernetes Native Java stack tailored for GraalVM & OpenJDK HotSpot
Cloud
Walk the talk
Energy Crisis
Retain & attract talented people
Assess
H2-2021 Release
Main Radar
Main Radar - Release H2 2021
Main Radar - Release H2 2022
Trial
Containers Infrastructure
In Foreign Radars
Status at Worldline
Production
Trial
Assess
On Hold
WorldWide maturity
Now to 1 year
1 to 3 years
3 to 6 years
6 years and more


Donuts At Home
[System]
Service Layer
[System]
Customer Service
[System]
Customer API
Java, Quarkus, Docker
JDBC
Customer DB
PostgreSQL
or
HTTPS
Billing Service
[System]
Billing Object Storage
S3
S3/HTTPS
Billing API
Java, Spring Boot, Docker
JDBC
Billing DB
MongoDB
or
Presentation Layer
[System]
Donuts At Home Admins
Customer
HTTPS
backoffice GUI
ReactJS
Back for Backoffice GUI
NodeJS, Typescript, Docker
GUI
[Provides Front end GUI]
ReactJS
Back for Front GUI
NodeJS, Typescript, Docker
Kafka
Event Processing Service
[System]
Event Processing
Kafka
Delivery Connector
Kafka Connect
Payment Connector
Kafka Connect
Delivery Information System
SIPS
online secure credit card payment
Shopping Service
[System]
Shopping API
Java, Spring Boot, Docker
JDBC
Shopping DB
PostgreSQL
or


Donuts At Home
[System]
Service Layer
[System]
Customer Service
[System]
Customer API
Java, Quarkus, Docker
JDBC
Customer DB
PostgreSQL
or
HTTPS
Billing Service
[System]
Billing Object Storage
S3
S3/HTTPS
Billing API
Java, SpringBoot, Docker
JDBC
Billing DB
MongoDB
or
Kafka
Event Processing Service
[System]
Delivery Connector
Kafka Connect
Event Processing
Kafka
Payment Connector
Kafka Connect
Kafka
Kafka
HTTPS
Delivery Information System
SIPS
online secure credit card payment
Shopping Service
[System]
Shopping API
Java, Spring Boot, Docker
JDBC
Shopping DB
PostgreSQL
or
Presentation Layer
[System]
backoffice GUI
ReactJS
Back for Backoffice GUI
NodeJS, Typescript, Docker
GUI
[Provides Front end GUI]
ReactJS
Back for Front GUI
NodeJS, Typescript, Docker
HTTPS
Donuts At Home Admins
Customer
HypeJS

Think Observability by Design!

It's a capability of the platform more than just a bunch of tools...



Probability

Impact	Low 1	Medium 2	High 3
Low 1			
Medium 2	- Response times are too high (> SLO)	- Unavailability of external systems - Unobservable platform	
High 3	- Unavailability of middleware - Database access error - SAN error - VLAN network error		

Some good practices



Risk analysis at different levels



Accept new points of view



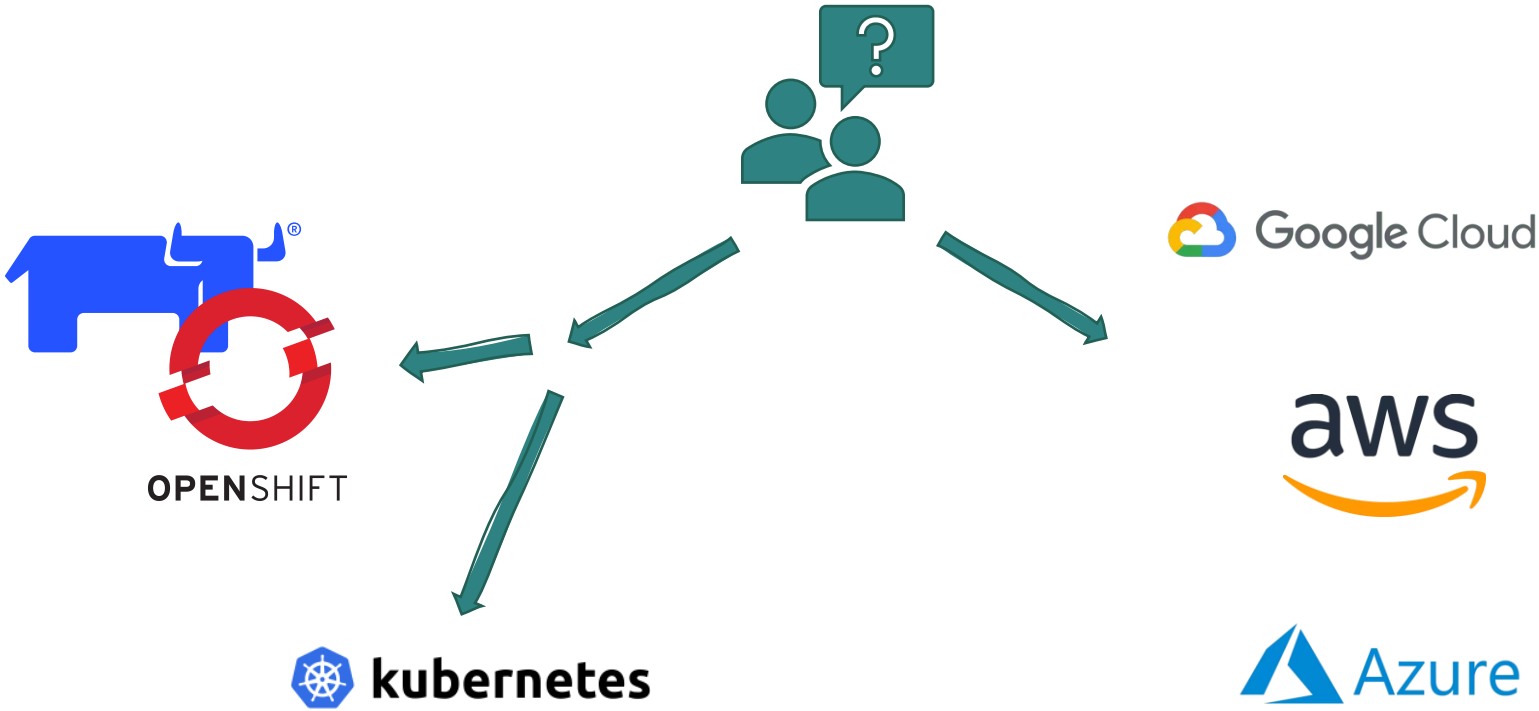
When to innovate?



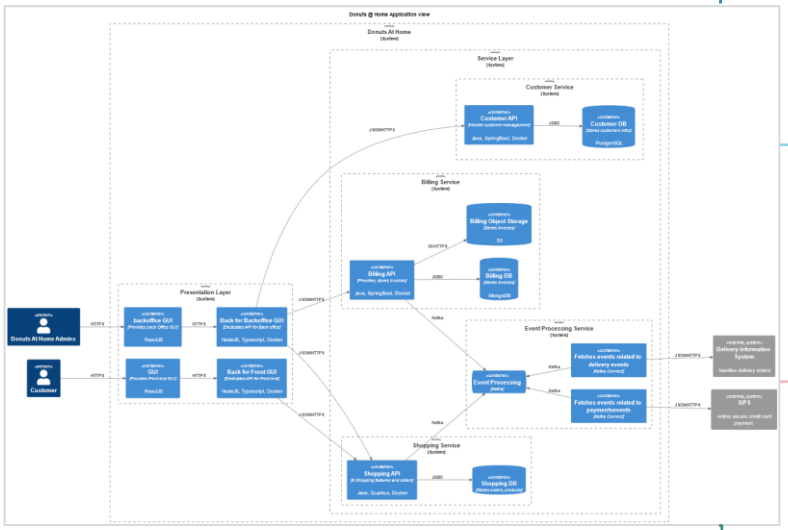
Organizational impacts

Infrastructure View

Private or Public Cloud?



How to **use the Cloud**?



Consume **external services**

Use **managed services**

Deploy **containers**

Deploy **virtual machines**

SaaS

Payment …

PaaS

APIM IAM Observability

CaaS

Spring Boot Quarkus MongoDB

IaaS

PostgreSQL Kafka

Architecture validation



Compliance with each view's requirements/constraints

- Involve Business, Devs, Ops, Finance, Experts
- Formalize commitment if necessary
- Possible validation steps

Regarding infrastructure

Verification of feasibility or technical hypotheses (POC) -
Non Functional Requirements -
Sizing elements -





Iterative work!

Conclusion



Attitude

- Step back
- Remain open-minded
- Share w/ all the stakeholders



Approach

- Formalise and trace
- Collaborate & iterate
- Align different views



Best practices & tools

- Blueprints & Patterns
- Technical roadmap
- Keep pragmatic

Thanks for your feedback!

https://bit.ly/vdb_archi



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